



BASE24-pos[®]

UK Credit Card Authorisation Interface Manual

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ACI documentation can be updated in two ways; new versions and revisions. New versions consist of a new document which replaces the old document. New versions incorporate all revisions since the previous version. Revisions consist of one or more pages to be inserted and/or replaced in the document by the user, each revised page is listed above.

Changes to this Version:

- Update manual to reflect upgrade of software to BASE24-pos Release 5.1 and inclusion of ICC support.
- General update of style and formatting.

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PREFACE

Intended Audience

This document is intended for users of the UK Credit Card Authorisation Interface who have a general knowledge of interface principles as well as BASE24 principles. Prior knowledge of the software is not essential.

Users should also consult the relevant BASE24 or Tandem manuals where appropriate.

This document can be used as an aid to training, but readers are advised to attend the relevant training course(s).

Location of Information

A list of section headings and related page references are given in the Contents page. The headings are function oriented.

Section 1—Introduction gives a brief overview of the UK Credit Card Authorisation Interface (UKCCI) and illustrates the flow of messages.

Section 2—Installation & Configuration describes file maintenance, XPNET configuration and BASE24 file configuration.

Section 3—Operations defines the NCS commands.

Appendix A—Log Messages.

Appendix B—Message Formats.

Appendix C—Sample Communications Environment and Configuration.

Appendix D—Standard Translation Table for the External Response Code and Internal Response Code.

Appendix E—Glossary which expands on the terms used in this manual.

User Comments Form.

Related Documentation

BASE24 Documentation

- XPNET Configuration Manual - Release 2.0
Publications Number : BMXP-NT000-01
- XPNET Network Environment Source File Manual - Release 2.0
Publications Number : BMXN-NT000-02
- XPNET Network Operators Manual - Release 2.0
Publications Number : BMXP-NT000-03
- BASE24 LCONF Manual - Release 5.1
Publications Number : BMBA-AE000-10
- BASE24 BASE Files Maintenance Manual - Release 5.1
Publications Number : BMBA-AE000-03
- BASE24-pos Introduction Manual - Release 5.1
Publications Number : BMPS-AE000-01
- BASE24-pos Files Maintenance Manual - Release 5.1
Publications Number : BMPS-AE000-02
- BASE24-pos Transaction Processing Manual - Release 5.1
Publications Number : BMPS-AE000-07
- BASE24-pos CRT Authorisation and CRT Admin. Manual - Release 5.1
Publications Number : BMPS-DP130-01

Third Party Documentation

- Standard 30, Specification for a Credit Authorisation Terminal Manual (1993), issued by APACS (Administration) Limited.

User Comments

At the back of this manual, you will find a form that can be used to comment on any aspect of this document.

We hope that you will be able to help by telling us what you like and what improvements you would like to see in future editions.

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1. INTRODUCTION

The UK Credit Card Authorisation Interface (UKCCI) allows institutions with a BASE24-pos system to acquire transactions for other card issuers who do not belong to the given institution's logical network. The Interface is an Issuer Host Interface and is referred to throughout this manual as the **Host Interface**.

All the card issuers conform to a standard specified in the Standard 30, Specification for a Credit Authorisation Terminal Manual (1993), issued by APACS (Administration) Limited, referred to as the **APACS30 Standard** throughout this manual.

The Host Interface provides an online message interface between the BASE24-pos system and an Issuer Host.

The Host Interface converts the POS Standard Internal Message (PSTM) to a Standard APACS Message (SAM) for routing to the Issuer Host, using the following files:

- Host Institution File (HIF)
- Host Retailer File (HRF)
- Host Terminal File (HTF)
- Host Configuration File (HCF).

Conversely, any messages from the Issuer Host are converted back to the PSTM format, for internal use by BASE24.

For more information on BASE24-pos, refer to the Introduction to BASE24-pos Manual and the BASE24-pos System Description Manual.

1.1.1 ICC Support

On receipt of a PSTM containing the ICC Status token, the first character of the Terminal Type field is set to indicate ICC Capability.

On receipt of a PSTM containing the ICC Request token, the standard fields of the external message are formatted and the UKCCI ICC module is called to format the additional ICC fields before sending the request to the Host.

In addition, the UKCCI ICC module is called to format the Card Details field for an ICC transaction.

On receipt of a response message from the Host containing ICC data, the UKCCI ICC module is called to retrieve the ICC data, add the ICC Response token and, if ICC Issuer Script Data is included in the response, add the ICC Script token to the PSTM before sending the response to Router/Auth.

Refer to the ACI Product Specification for APACS ICC, UK Credit Card Authorisation Interface, for full details of ICC support.

Figure 1-1 shows the Host Interface in relation to other processes in the BASE24-pos system.

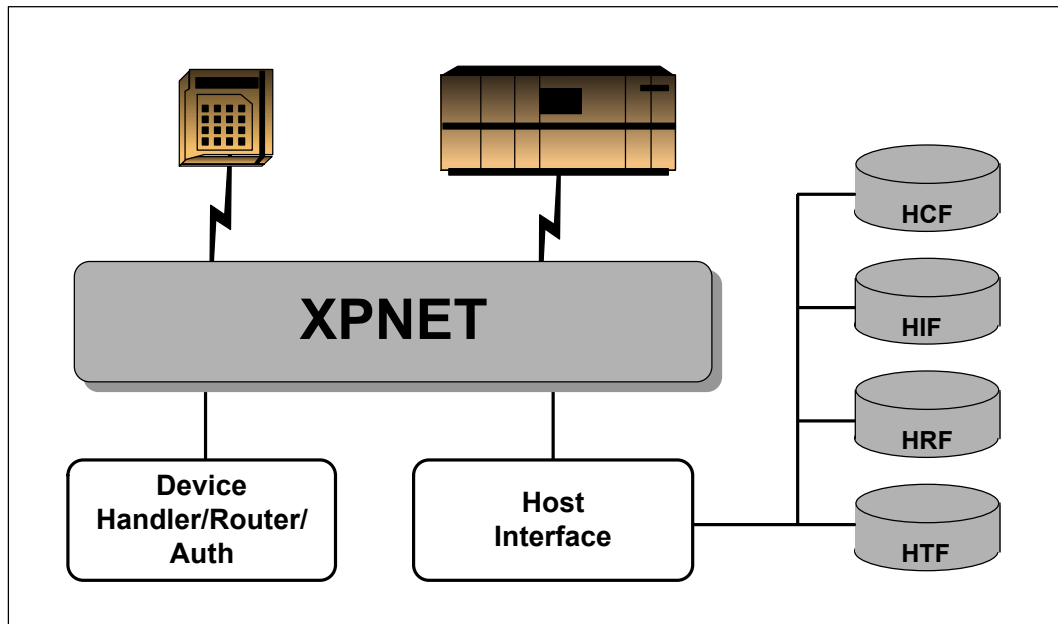


Figure 1-1 Relationship to other BASE24-pos Processes

The Host Interface supports:

- APACS30 Authorisation Request
- APACS30 Authorisation Response
- APACS30 Authorisation Confirmation
- APACS30 Hold Formats.

The types of transactions are:

- First try card-swiped Sale/Cheque Guarantee/Cash Advance
- Card-keyed Sale/Cheque Guarantee/Cash Advance.

For details of the flow of messages, refer to section 1.2.

Note that the following functions are not supported:

- Floor limits processing
- An alternate phone number or network user address
- Reversal message processing between the front-end and back-end system
- Message MACing
- PIN processing.

1.2 MESSAGE FLOW

This section provides details of the following types of message flow:

- Authorisation Request
- Authorisation Response
- Authorisation Confirmation
- Hold Message.

These messages are used by the Host Interface process to communicate with the external host.

1.2.1 Authorisation Request/Response

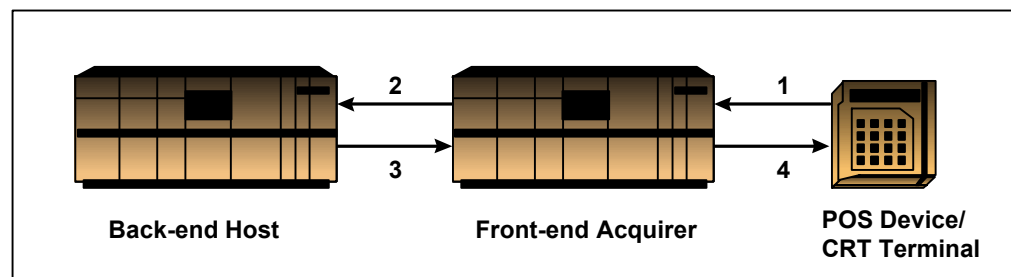


Figure 1-2 Authorisation Request/Response Message Flow

The following explains the flow of the authorisation request/response message:

1. A request is sent from a POS device or CRT terminal to a front-end acquirer.
2. The request is forwarded to the back-end host via the Host Interface.
3. The back-end host performs authorisation and returns a response to the front-end acquirer, via the Host Interface.
4. The front-end acquirer forwards the response message to the POS device or CRT terminal.

1.2.2 Authorisation Request/Response/Confirmation

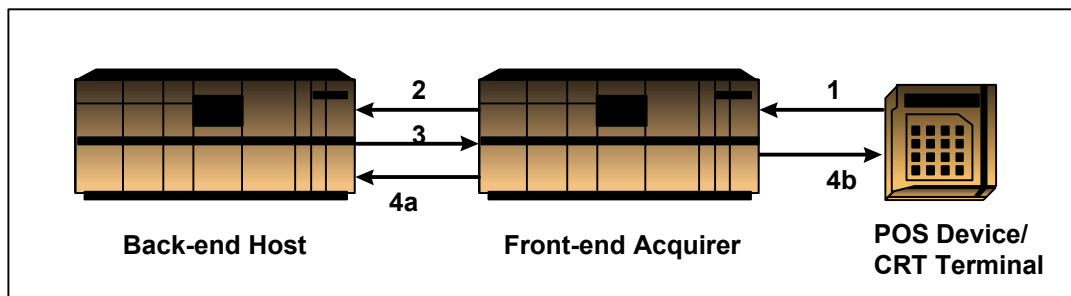


Figure 1-3 Authorisation Request/Response/Confirmation Message Flow

The following explains the flow of the authorisation request/response/confirmation message:

1. A request is sent from a POS device or CRT terminal to a front-end acquirer.
2. The request is forwarded to the back-end host via the Host Interface.
3. The back-end host performs authorisation and returns a response to the front-end acquirer, via the Host Interface.
- 4a. The Host Interface generates a confirmation message to the back-end issuer host.
- 4b. The front-end acquirer forwards the response message to the POS device or CRT terminal.

1.2.3 Authorisation Request/Hold/Response

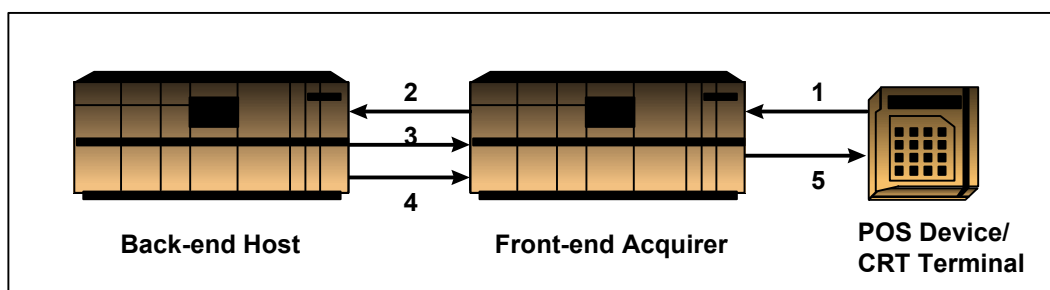


Figure 1-4 Authorisation Request/Hold/Response Message Flow

The following explains the flow of the authorisation request/hold/response message:

1. A request is sent from a POS device or CRT terminal to a front-end acquirer.
2. The request is forwarded to the back-end host via the Host Interface.
3. The back-end host sends a hold message to the front-end acquirer Host Interface to request that the timer be extended by 120 seconds.
4. The back-end host performs authorisation and returns a response to the front-end acquirer, via the Host Interface.
5. The front-end acquirer forwards the response message to the POS device or CRT terminal.

2. INSTALLATION AND CONFIGURATION

2.1 FILE MAINTENANCE

The UK Credit Card Authorisation Interface files can be accessed via the BASE Product Menu from the Virtual Menu of the BASE24 AFT subsystem.

The files are:

- Host Institution File (HIF)
- Host Retailer File (HRF)
- Host Terminal File (HTF)
- Host Configuration File (HCF).

Details on the above files are provided in the sections that follow.

To access the UK Credit Card Authorisation Interface files, refer to the following manuals:

- BASE24 BASE Files Maintenance Manual
- BASE24-pos CRT Authorisation and CRT Admin. Manual.

2.1.1 Host Institution File (HIF)

The Host Institution File (HIF) defines external issuer details and contains one record for each external issuer to which the Host Interface connects.

The AFT File Maintenance on the HIF is logged to the OMF with a file application code of '95' in the OMF record.

The primary key is the external issuer Financial Institution Identifier (FIID).

The fields displayed on the HIF screen (Figure 2-1) are defined on the following pages.

BASE24-BASE		HOST INSTITUTION FILE		ICC3	04/01/11	06:43	01 OF 01
HOST DPC/FIID :							
DIAL IND	: 4	TERM ID :	RETAILER ID :				
TERM TYPE :	INPUT TERM ID FORMAT : A APACS40						
REF PHONE RANGE	FROM : 0	TO : 3	LRC REQUIRED : 0				
HOLD MESSAGE	: 0	NOT SUPPORTED					KEYED READ
DISCONNECT MESSAGE	: 0	NOT SUPPORTED					START SENTINEL : 3B 3B
							UNIT SEPARATOR : 3D 3D
							END SENTINEL : 3F 3F
EXTERNAL RESPONSE CODE TABLE							
EXT-RSP	MESSAGE	INT-RSP	EXT-RSP	MESSAGE	INT-RSP		
***** BASE24 *****							
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:					
	F12-HELP						

Figure 2-1 Host Institution File (HIF) Screen

HOST DPC/FIID

4 (AN) M

The FIID of the financial institution which issued the card. The FIID must be a unique identifier within the logical network. A blank identifier is not permitted.

Default Value: None.

Related Fields: IDF.FIID

DIAL IND

1 (N) M

A dial indicator used in the external message.

Default Value: 4.

Related Fields: None.

TERM ID**8 (AN) M**

This field specifies the substitution method to be used in forming the External Terminal Id. There are four substitution methods which can be specified, depending on the value of this field:

- **No Substitution**

Value: '*****'

This specifies that the External Terminal Id is to be formed from the first eight bytes of the Internal Terminal Id (see related Input Term Id Format field).

- **Modified Generic Substitution**

Value: contains 1-7 characters with trailing asterisks; for example, '42*****'.

This specifies that the External Terminal Id is to be formed using the leading characters from this field with the trailing asterisks replaced by the least significant characters from the Internal Terminal Id (see related Input Term Id Format field).

- **Station Generic Substitution**

Value: 'STATION '

This specifies that each station used by the Data Processing Centre (DPC) has an associated External Terminal Id which is used as the External Terminal Id for the message (see the Station Configuration Record in the HTF).

- **Generic Substitution**

Value: contains any string excluding those which conform to the other substitution methods.

This specifies that the value in this field is to be used as the External Terminal Id for the message.

Default Value: None.

Related Fields: Input Term Id Format, Terminal Substitution Record in HTF, Station Configuration Record in HTF.

RETAILER ID**15 (AN) M**

This field specifies the substitution method to be used in forming the External Merchant Id. There are three substitution methods which can be specified, depending on the value of this field:

- **No Substitution**

Value: '*****'

This specifies that the External Merchant Id is to be formed from the first 15 bytes of the Internal Retailer Id.

- **Exact Substitution**

Value: Spaces.

This specifies that an exact Retailer Id is to be used in the external message. The exact Retailer Id is retrieved from the HRF using the Host FIID and the Internal Retailer Id as the key (see the External Merchant Id in the HRF).

- **Generic Substitution**

Value: contains any string excluding spaces and asterisks.

This specifies that the value in this field is to be used as the Merchant Id for the external message.

Default Value: None.

Related Fields: Internal Retailer Id, External Merchant Id in HRF.

TERM TYPE**4 (N) M**

This field specifies the value of the Terminal Type field for the external message. The valid characters are:

- 1st character is 2 or 4
- 2nd character is 0-5
- 3rd character is 0-7
- 4th character is 0-9.

Default Value: None.

Related Fields: None.

For more information, refer to the APACS30 Standard.

INPUT TERM ID FORMAT**1 (A)**

This field defines the format of the Internal Terminal Id for the Host Interface. This field is only relevant when 'No Substitution' or 'Modified Generic Substitution' methods have been specified in the Term Id field.

The valid values are:

- | | |
|---|---|
| A | The Internal Terminal Id is assumed to be in APACS30 format—the 'No Substitution' or 'Modified Generic Substitution' method will function as specified in the Term Id field. |
| N | The Internal Terminal Id is assumed not to be in APACS30 format—the Host FIID and the Internal Terminal Id are used as the key to retrieve the Terminal Substitution Record from the HTF; the External Term Id field of the record is used in the 'No Substitution' or 'Modified Generic Substitution' methods in place of the Internal Terminal Id. |
| M | The Internal Terminal Id is assumed to be either of the above formats (A or N)—the Host Interface will determine whether or not the format of the Internal Terminal Id is in APACS30 format, and acts in accordance with the descriptions above. The Host Interface will assume the Internal Terminal Id is APACS30 format if the field consists of eight digits. |

Default Value: None.

Related Fields: Term Id, Terminal Substitution Record in the HTF.

REF PHONE RANGE

The Referral Phone Range consists of two fields:

- Ref Phone Range From
- Ref Phone Range To

These fields specify the range of substitution methods that can be used to determine the referral telephone number to be passed back to the originating terminal. The four substitution methods are:

- **No Substitution**
The Referral Phone Number from the external message is returned, if one exists.
- **Exact Substitution**
The Referral Phone Number from the HRF record is returned, if one exists (see Referral Phone Number in the HRF).
- **Generic Substitution**
The Referral Phone Number from the HCF record for the current DPC is returned, if one exists.
- **Default Substitution**
The Referral Phone Number present in the original internal request is returned.

The Ref Phone Number From field specifies the substitution method which should be used first in attempting to resolve the Referral Phone Number.

The Ref Phone Number To field specifies the last substitution method which should be used in attempting to resolve the Referral Phone Number.

The Ref Phone Number To value must be greater than or equal to the Ref Phone Range From field.

The Host Interface will apply each substitution method specified within the range until it has obtained the Referral Phone Number to be returned to the originating terminal. Where the Host Interface has applied the substitution method specified in the Ref Phone Number To field and is still unable to resolve the Referral Phone Number, the Default Substitution method is used.

Where the external response contains a Referral Queue Number of n digits and the No Substitution method is used, the Host Interface will attempt to form a Referral Phone Number. This is done by taking the generic number in the HCF (if it exists) and replacing the last n digits of the generic number with the Referral Queue Number.

Where the Referral Phone Number field of the external response contains a single Unit Separator (US) and the No Substitution method is used, the Host Interface will attempt to form a Referral Phone Number. This is done by taking the generic number in the HCF (if it exists).

Related Fields: Referral Phone Number in external response, Referral Phone Number in HRF, Referral Phone Number in POS Product Data of HCF and Referral Phone Number in the internal message.

REF PHONE RANGE FROM**1 (N) M**

The valid values range from 0-3.

Default Value: 0

Related Fields: Ref Phone Range To.

REF PHONE RANGE TO**1 (N) M**

The value specified must be greater than or equal to the Ref Phone Range From field. The valid values range from 0-3.

Default Value: 3

Related Fields: Ref Phone Range From.

LRC REQUIRED**1 (N) M**

This code specifies whether the LRC of TRACK2 of the internal message is to be included in the Card Details field of the external message for cards which have been magnetically read. The valid values are:

0 Omit LRC (Default)

1 Include LRC.

Related Fields: None.

HOLD MESSAGE**1 (N) M**

This field specifies whether the Host Interface will support hold messages from the issuer. If the field is set to support hold messages, the Host Interface will extend an outstanding request time by 120 seconds for each hold message received from the issuer for that request. The valid values are:

0 Hold message is *not supported* (Default)

1 Hold message is *supported*.

The above text in italics is displayed automatically for the given value.

Related Fields: None.

DISCONNECT MESSAGE**1 (N) M**

This field specifies whether the Host Interface will disconnect the line (X25 protocol only) on normal or abnormal completion of a transaction. Where the DPC is multi-threaded, the disconnect will only be initiated when the line (station) has acquiesced. The valid values are:

0 Disconnect message is *not supported* (Default)

1 Disconnect message is *supported*.

The above text in italics is displayed automatically for the given value.

Related Fields: DPC type in HCF.

CARD DETAILS TABLE**Occurs 2 times**

There are two entries in the table:

- The first entry contains information relating to card-read transactions.
- The second entry contains information relating to card-keyed transactions.

This table specifies replacement characters for Start Sentinel, Unit Separator and End Sentinel. The Start Sentinel, Unit Separator and End Sentinel, if found in TRACK2 of the internal message, will be replaced by the values specified in the table when building the Card Details field of the external request.

The replacement characters are specified in hexadecimal notation and can be any character in the range 01-FF. The null character (00) is reserved to indicate character stripping.

START SENTINEL**2 (AN) M**

This field contains the hexadecimal representation of the replacement character for the Start Sentinel. The valid values are in the range 01-FF. The null character (00) will strip the Start Sentinel when building the Card Details field of the external request.

Default Value: 3B (;)

Related Fields: None.

UNIT SEPARATOR**2 (AN) M**

This field contains the hexadecimal representation of the replacement character for the Unit Separator. The valid values are in the range 01-FF. The null character (00) will strip the Unit Separator when building the Card Details field of the external request.

Default Value: 3D (=)

Related Fields: None.

END SENTINEL**2 (AN) M**

This field contains the hexadecimal representation of the replacement character for the End Sentinel. The valid values are in the range 01-FF. The null character (00) will strip the End Sentinel when building the Card Details field of the external request.

Default Value: 3F (?)

Related Fields: None.

EXTERNAL RESPONSE CODE TABLE

There are 16 entries in the External Response Code Table. Each entry contains an external response code, a 16 byte message and an associated Internal Response Code. Unused entries are left blank. This table is optional.

External Response Codes are translated to Internal Response Codes in four ways, the first two of which involve this table:

- 1 External Response Code and Message Translation by Issuer—
An attempt is made to match the Response Code and the first 16 bytes of the Message field in the External Message to an entry in the External Response Code Table for the issuer. If a matching entry is found, the Internal Response Code for that entry is used as the Internal Response Code.
- 2 External Response Code Translation by Issuer—
If the above method did not result in a successful translation of the External Response Code, an attempt is made to match the External Response Code in the External Message to an entry in the External Response Code Table, where the Message field of the entry in the table is spaces.
- 3 External Response Code Translation—
If neither of the above methods resulted in a successful translation of the External Response Code, an attempt is made to translate the External Response Code using the Standard Translation Table (see Appendix D).
- 4 Default External Response Code Translation—
If none of the above methods resulted in a successful translation of the External Response Code, a Default Internal Response Code of 'Refer to Issuer' (101) is used.

EXT-RSP

2 (AN)

This field contains the External Response Code. If this field is not blank, there must be a non-blank value in the corresponding Int-Rsp field.

Default Value: Spaces
Related Fields: INT-RSP.

MESSAGE

2 (AN)

When used in External Response Code and Message Translation, this field must contain values expected in the first 16 bytes of the Message field of the External Response associated with the External Response Code.

When used in External Response Code Translation, this field must be spaces. If this field is not spaces, the Ext-Rsp and Int-Rsp fields must each contain a non-blank value.

Default Value: Spaces
Related Fields: EXT-RSP; INT-RSP.

INT-RSP

2 (AN)

This field is an Internal Response Code. If this field is not blank, there must be a non-blank value in the corresponding Ext-Rsp field.

Default Value: Spaces
Related Fields: EXT-RSP.

2.1.2 Host Retailer File (HRF)

The Host Retailer File (HRF) contains one record for every external issuer and acquiring retailer combination. These records are only required if 'Exact Retailer Substitution' and/or 'Exact Referral Phone Number Substitution' are specified in the HIF (see section 2.1.1).

The HRF defines:

- The Exact Substitute Merchant Identifier.
- The Exact Substitute Referral Phone Number.

The AFT File Maintenance on the HRF is logged to the OMF with a file application code of '97' in the OMF record.

The primary key is the external issuer FIID and the acquiring retailer identifier.

The fields displayed on the HRF screen (Figure 2-2) are defined on the following pages.

BASE24-BASE HOST RETAILER FILE		ICC3	04/01/11 06:44 01 OF 01
HOST DPC/FIID:	RETAILER ID:		
EXTERNAL MERCHANT ID :			
REFERRAL PHONE NO. :			
***** BASE24 *****			
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
F12-HELP			

Figure 2-2 Host Retailer File (HRF) Screen

HOST DPC/FIID

4 (AN) M

This field contains a unique identifier for an issuer host. The Host FIID and the Retailer Id form the primary key for the file. Note that spaces are not permitted in this field.

Default Value: None.

Related Fields: Host FIID in HIF.

RETAILER ID**9 (AN) M**

This field contains a unique identifier for the front-end retailer. The Retailer Id and the Host FIID form the primary key for the file.

Default Value: None.

Related Fields: RETAILER ID in PTDF.

EXTERNAL MERCHANT ID**15 (AN)**

Optionally, this field may contain an Exact Substitute Merchant Identifier. If the Referral Phone Number field is blank, this field is mandatory.

Default Value: None.

Related Fields: RETAILER ID in HIF.

REFERRAL PHONE NO**20 (N)**

Optionally, this field may contain an Exact Substitute Referral Phone Number. If the External Merchant Id field is blank, this field is mandatory.

Default Value: None.

Related Fields: REF PHONE RANGE in HIF.

2.1.3 Host Terminal File (HTF)

The Host Terminal File (HTF) contains three types of record:

- **Online Terminal Record (OTR)**

The OTR stores the message sequence number for each External Terminal Id it uses. The record is created automatically when required by the Host Interface.

The key to this type of record consists of:

- ⇒ Host FIID
- ⇒ Internal Id of asterisks
- ⇒ External Terminal Id.

- **Terminal Substitution Record (TSR)**

The TSR stores a terminal identifier which is used by the 'No Terminal Substitution' and 'Modified Generic Terminal Substitution' methods in place of the Terminal Id in the External Message. This occurs when the Input Term Id Format field in the HIF is set to M or N, and the Terminal Id in the Internal Message is not in APACS30 format.

This type of record is not required if the 'Generic Terminal Substitution' or 'Station Generic Terminal Substitution' methods are used.

The key to this type of record consists of:

- ⇒ Host FIID
- ⇒ Internal Terminal Id
- ⇒ External Terminal Id.

- **Station Configuration Record (SCR)**

The SCR stores an External Terminal Id for each station which belongs to a DPC that uses 'Station Generic Terminal Substitution' when the Term Id in the HIF is set to STATION.

The key to this type of record consists of:

- ⇒ Host FIID of asterisks
- ⇒ BASE24 Station Symbolic Name
- ⇒ External Terminal Id.

The AFT File Maintenance on the HTF is logged to the OMF with a file application code of '96' in the OMF record.

The fields displayed on the HTF screen (Figure 2-3) are defined on the following pages.

```

BASE24-BASE  HOST TERMINAL FILE      ICC3      04/01/11  06:43  01 OF 01
HOST DPC/FIID: █
INTERNAL ID   :                      EXTERNAL TERM ID:

                                TERMINAL SUBSTITUTION RECORD

MESSAGE NUMBER: 0000

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 2-3 Host Terminal File (HTF) Screen

HOST DPC/FIID

4 (AN) M

This field contains a unique identifier for an issuer host. The Host FIID, Internal Id and External Term Id fields form the key to the file. Note that spaces are not permitted in this field.

A Host FIID of asterisks indicates that this is a Station Configuration Record (SCR) and the SCR is displayed.

Default Value: None.

Related Fields: Host FIID in HIF.

INTERNAL ID

16 (AN) M

- For an Online Terminal Record, this field should contain asterisks and the Online Record is displayed.
- For a Terminal Substitution Record, this field should contain the Internal Terminal Id and the Terminal Substitution Record is displayed.
- For a Station Configuration Record, this field should contain the BASE24 Station Symbolic Name.

Default Value: None.

Related Fields: TERM ID and INPUT TERM ID FORMAT in HIF.

EXTERNAL TERM ID**8 (AN) M**

This field contains the Terminal Identifier used by the different substitution methods.

Default Value: None.

Related Fields: TERM ID and INPUT TERM ID FORMAT in HIF.

TERMINAL SUBSTITUTION RECORD**MESSAGE NUMBER****(display only)**

This is a display only field. Each time this field is used by the Host Interface as a message sequence number in an external message, its value is incremented by 1. The valid values are in the range 0000-9999. The counter display is rolled over to 0000 after it has reached 9999. For an ADD operation, the value is set to 0000.

2.1.4 Host Configuration File (HCF)

The Host Interface uses a subset of the fields in the Host Configuration File (HCF):

- DPC Number
- HISF Name
- Performance Period
- DPC Type
- ACK to DPC
- Maximum Outstanding Requests Outbound
- Data Prefix Characters
- Station Table
- POS Product Data Referral Phone Number
- POS Product Data Timer Limits Outbound Limit
- POS Product Data Timer Limits Queue Subtract.

The above fields, with the exception of the ACK to DPC and Data Prefix Characters, are fully described in the BASE24 BASE File Maintenance Manual.

The ACK to DPC and Data Prefix Characters fields are described below:

ACK TO DPC

1 (N) M

This field specifies whether the Host Interface will support confirmation messages to the issuer. Valid values are:

- 0 Confirmation message is not supported (Default)
- 1 Confirmation message is supported.

Related Fields: None.

DATA PREFIX CHARACTERS

8 x 2 (AN)

This first Data Prefix Character should be set to a start-of-text (02 in hexadecimal).

Default Value: Spaces

Related Fields: None.

2.2 XPNET CONFIGURATION

This section gives examples of the Network Environment File Source (NEFS) definitions which are required to install the UKCCI using a PSS X25 line.

The following definitions are detailed in the sections below:

- Process
- Device
- Station
- Line.

The definitions are required for the Host Interface and communications port.

For more information, refer to the Network Environment Source File Manual.

2.2.1 Process Definitions

```

PIA^HAPCI                      =1:
OBJECT                          '$DATA.BA51HAPC.HAPC',
LIBRARY                         '$SYSTEM.SPANNET.SKELB',
CPU                             2,
BOJ,
MAX CORE QUEUE                  100,
SYMBOLIC USAGE,
LOGICAL NETWORK IS              APA1,
PPD NAME IS                     '$HAP1',
PRIORITY                        160;

```

2.2.2 Device Definitions

```

PSS^DEV                        =2:
DEVICE TYPE                     0,
TRANSMIT                        128,
RECEIVE                         128,
TRANSMISSION NUMBER LENGTH     0,
ADDRESS CHARACTER LENGTH        2,
UNIT ADDRESS POSITION            2,
NORMAL SELECT,
XPNET RESIDENT SSCP NO;

```

2.2.3 Station Definitions

<i>SIA^HAPCI</i>	<i>=1:</i>
<i>DEVICE PSS^DEV,</i>	
<i>DESTINATION PROCESS IS</i>	<i>PIA^HAPCI,</i>
<i>LOGICAL NETWORK IS</i>	<i>APA1,</i>
<i>OUTPUT INTERVAL</i>	<i>0,</i>
<i>MAX CORE QUEUE</i>	<i>100,</i>
<i>RETRY</i>	<i>0,</i>
<i>TERMINAL TYPE</i>	<i>0,</i>
<i>NO LOGICAL ACK,</i>	
<i>AUTOSTART PRIORITY</i>	<i>128,</i>
<i>WARMSTART PRIORITY</i>	<i>128,</i>
<i>SENSE STATUS QUEUE COUNT</i>	<i>0;</i>

2.2.4 Line Definitions

<i>LIA^HAPCI</i>	<i>=1:</i>
<i>DEVICE ID</i>	<i>\$PSS.#HERTZ,</i>
<i>X25,</i>	
<i>STATIONS</i>	<i>SIAHAPCI,</i>
<i>SYNC,</i>	
<i>TIMER INTERVAL</i>	<i>1 SECOND,</i>
<i>TIMEOUT</i>	<i>1 SECOND,</i>
<i>RETRY</i>	<i>1,</i>
<i>CONTROL CHARACTER OCCURS</i>	<i>2,</i>
<i>UNIT CHARACTER OCCURS</i>	<i>2,</i>
<i>POLL INTERVAL</i>	<i>1 SECOND,</i>
<i>OUTPUT BIAS</i>	<i>3,</i>
<i>ERROR ANALYSIS INTERVAL</i>	<i>30 SECONDS,</i>
<i>WAIT FOR CALL,</i>	
<i>MAKE CALL,</i>	
<i>END TO END PROTOCOL</i>	<i>0,</i>
<i>RECALL DELAY</i>	<i>2,</i>
<i>ATIMER</i>	<i>45,</i>
<i>FULL DUPLEX;</i>	

2.3 BASE24 FILE CONFIGURATION

This section specifies the assigns and parameters that must be present in the Logical Network Configuration File (LCONF) for the Host Interface.

2.3.1 LCONF Assigns

The following assigns must be present in the LCONF:

HCF	Specifies the location of the Host Configuration File. This file is used by all Host Interfaces.
HIF	Specifies the location of the Host Institution File. This file is used by the UKCCI.
HRF	Specifies the location of the Host Retailer File. This file is used by the UKCCI.
HTF	Specifies the location of the Host Terminal File. This file is used by the UKCCI.
LMCF	Specifies the location of the Log Message Configuration File. This file is used by all BASE24 modules.
PLF	Optionally, specifies the location of the PCS Log File. This file is used by the PCS System.
SPANCTL	Specifies the location of the SPAN Control File. This file is used to locate the DESTMAP file.

2.3.2 LCONF Parameters

The following parameters must be present in the LCONF:

LOGICAL-NET	Specifies the Logical Network.
FIIDS-WITH-DPC-HIF-KEY	Optionally, specifies the FIIDs associated with a DPC.
DISC-STATUS-TIMER	Optionally, specifies the time limit for inquiring on a disconnected station.
POS-LN-CUTOVER	Specifies the POS Logical Network Cutover Time.
PS-SEG	Specifies the identifier for the AFT Segment.

ACI Limited

3. OPERATIONS

3.1 NCS COMMANDS

The Host Interface process routinely receives a set of commands which cause it to perform specialised types of activities.

The list of NCS commands include:

- LOGON
- LOGOFF
- PERFORMANCE
- STATUS
- TRACE
- WARMBOOT

Details of the commands are provided below.

3.1.1 LOGON Command

Originator Network Control Supervisor screen.

Message form LOGON [ALL] (for all DPCs)
LOGON [nnnn] (for a particular DPC number)

Description This command is used to mark one or more DPCs up.

If the optional parameter 'ALL' is specified in the message, the above actions are taken for all DPCs processed by the Host Interface process. However, if a DPC number is specified, the action is taken for that DPC only.

3.1.2 LOGOFF Command

Originator Network Control Supervisor screen.

Message form LOGOFF [ALL] (for all DPCs)
LOGOFF [nnnn] (for a particular DPC number)

Description This command is used to mark one or more DPCs down.

If the optional parameter 'ALL' is specified in the message, the above actions are taken for all DPCs processed by the Host Interface process. However, if a DPC number is specified, the action is taken for that DPC only.

3.1.3 PERFORMANCE Command

Originator Network Control Supervisor screen.

Message form PERFORMANCE [ALL] (for all DPCs)
PERFORMANCE [nnnn] (for a particular DPC number)

Description This command requests performance statistics from the Host Interface process. The Host Interface tracks performance statistics for each DPC it handles. These statistics are gathered on an interval basis, as specified by the interval length in the PERFORMANCE LIMIT field in the HCF. Statistics are available for the most current four intervals.

The Host Interface process logs the performance statistics. If the optional parameter 'ALL' is specified in the message, the Host Interface process logs statistics for all DPCs it processes. However, if a DPC number is specified, the Host Interface process logs statistics for that particular DPC only.

The statistics are formatted into table form, as follows:

<i>DPC NUMBER: nnn</i>		<i>MINUTES IN PERIOD: nnn</i>		
	<i>RQSTS</i>	<i>TIME OUT</i>	<i>PERCENT</i>	<i>RESP TIME</i>
<i>CURRENT</i>	<i>nnnn</i>	<i>nnnn</i>	<i>nnn%</i>	<i>nnn.nn</i>
<i>PERIOD 01</i>	<i>nnnn</i>	<i>nnnn</i>	<i>nnn%</i>	<i>nnn.nn</i>
<i>PERIOD 02</i>	<i>nnnn</i>	<i>nnnn</i>	<i>nnn%</i>	<i>nnn.nn</i>
<i>PERIOD 03</i>	<i>nnnn</i>	<i>nnnn</i>	<i>nnn%</i>	<i>nnn.nn</i>
<i>PERIOD 04</i>	<i>nnnn</i>	<i>nnnn</i>	<i>nnn%</i>	<i>nnn.nn</i>

where 'n' specifies a numeric entry.

.

3.1.4 STATUS Command

Originator	Network Control Supervisor screen.
Message form	STATUS [ALL] (for all DPCs) STATUS [nnnn] (for a particular DPC number)
Description	This command requests status information for the DPC(s) handled by the Host Interface process. The Host Interface sends the requested status information to its log.

If the optional parameter 'ALL' is specified in the message, the Host Interface process logs the requested status information for all DPCs it processes. However, if a DPC number is specified, the Host Interface process logs the requested status information for that particular DPC only.

The status information is formatted into table form, as follows:

```
DPC STATUS SUMMARY--BASE24 RELEASE 5.1 yy/mm/dd hh:mm:ss
REQUESTS--PENDING: nnnn ALLOWED: nnnn DEPTH: nnnn
IDX* NUM* STAT* FIID* ACK* STA* SUP* SUS* MAX* THRD
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
nnn nnn xx xxxx x nn nn nnn nnn xxxx
```

where:

IDX	specifies the DPC index.
NUM	specifies the DPC number.
STAT	specifies the status of the DPC. The valid values are UP (Up) or DN (Down).
FIID	specifies the Host FIID of the DPC.
ACK	specifies whether a confirmation is to be sent to the Issuer Host. The valid values are Y (Yes) or N (No).
STA	specifies the number of stations.
SUP	specifies the number of stations that are up.
SUS	specifies the number of requests currently suspended.
MAX	specifies the maximum number of suspended requests that were handled by the DPC since the DPC was up.
THRD	specifies whether the DPC is multi-threaded (MULT) or single-threaded (SNGL).
'n'	specifies a numeric entry.
'x'	specifies an alphanumeric entry.

3.1.5 TRACE Command

Originator	Network Control Supervisor screen.	
Message form	TRACE ON	(turns the trace function on)
	TRACE OFF	(turns the trace function off)
Description	This command is a debugging aid. If TRACE is set to ON, extra log messages will be displayed on the screen. If TRACE is set to OFF, no extra log messages will be displayed.	

3.1.6 WARMBOOT Command

Originator	Network Control Supervisor screen.	
Message form	9501 WARMBOOT	
Description	This command warmboots the Host Interface process. This forces the Host Interface process to reinitialise its internal tables by re-reading the following files: ⇒ LCONF ⇒ LMCF ⇒ HCF ⇒ HIF. Should the Host Interface process fail to complete the warmboot, the Host Interface will resume operation with its current internal tables intact.	

A. LOG MESSAGES

A.1 FORMAT OF MESSAGES

Log messages are generated to report an error condition or information. Error messages may require corrective action before the Host Interface can function properly.

The standard format for BASE24 log messages is:

**YY/MM/DD HH:MM:SS (X) <PROCESS ID> <OBJECT> MMMM RRRR
<TEXT>**

where:

- 'YY/MM/DD' specifies the date the message was sent.
- 'HH:MM:SS' specifies the time the message was sent.
- '(X)' specifies the severity code where:

(0) or (I)	Status (Informational)
(1) or (W)	Warning
(2) or (E)	Error
(3) or (C)	Critical
- '<PROCESS ID>' specifies the symbolic process name.
- '<OBJECT>' specifies a 16-byte string defined and passed by the process.
- 'MMMM' specifies the message number.
- 'RRRR' specifies the routing code.
- '<TEXT>' specifies the text of the message.

For example:

**97/03/21 12:00:00 (2) P1A^HAPC 2010 0002
UNABLE TO DELETE SUSPENDED RECORD**

To avoid the repetition of showing the date, time, process identifier, object and routing code for each message; only the message number, severity code and the text of the message are documented. Therefore, the display format of log messages adopted for documentation is:

MMMM (X) <TEXT>

For example:

2010 (2) UNABLE TO DELETE SUSPENDED RECORD

For each log message, additional information will be given:

- the cause or reason the message is displayed
- the remedy or solution for the message, if relevant.

The following support/service personnel may need to be contacted:

- Operations Personnel
In-house operations staff who are responsible for initiating fault action for a particular problem.
- HELP24
ACI Limited personnel who are responsible for solving customer queries and problems.

A.2 UKCCI LOG MESSAGES

Log Messages added to the standard BASE24-pos messages to assist in the diagnosis of UK Credit Card Authorisation Interface error conditions are documented in this section.

2001	(2)	UNABLE TO PROCESS PSTM REQUEST. UNKNOWN TRAN CODE \
-------------	------------	--

where: 'W' specifies the transaction code from the PSTM request.

CAUSE: The PSTM request contains a transaction code that is not recognised by the Host Interface.

REMEDY: Check the IDF transaction table. Contact Operations Personnel if the problem persists.

2004	(2)	UNABLE TO PROCESS PSTM REQUEST. UNKNOWN FIID \\\ FOR DPC #
-------------	------------	---

where: '\\\' specifies the FIID from the PSTM request.
'#' specifies the DPC number from the PSTM request.

CAUSE: The PSTM request from the Acquirer contains an invalid FIID. There is no record for the FIID in the HIF.

REMEDY: If the FIID is valid, create a record in the HIF for the FIID and warmboot the process.

2005	(2)	UNABLE TO PROCESS PSTM REQUEST. UNKNOWN DPC #
-------------	------------	--

where: '#' specifies the DPC number from the PSTM request.

CAUSE: The PSTM request from the Acquirer contains an invalid DPC number. There is no record for the DPC number in the HCF.

REMEDY: Create a record for the DPC number in the HIF and warmboot the process.

2006	(2)	UNABLE TO PROCESS PSTM REQUEST. EXTERNAL TERMINAL ID FOR FIID \\\ AND PSTM TERMINAL ID \? NOT FOUND IN \F
-------------	------------	--

where: '\\\' specifies the FIID from the PSTM request.
'\'?' specifies the Terminal Id from the PSTM request.
'F' specifies the HTF file name.

CAUSE: The PSTM request from the Acquirer contains the missing details specified. A record was not found in the HTF.

REMEDY: Create the appropriate HTF record and warmboot the process.

2007	(2)	UNABLE TO PROCESS PSTM REQUEST. STATION TERM ID FOR \S NOT FOUND IN \F
-------------	------------	---

where: 'S' specifies the Station symbolic name from the PSTM request.
'F' specifies the HTF file name.

CAUSE: The PSTM request from the Acquirer contains an invalid Station Identifier. A Station Configuration Record (SCR) record was not found in the HTF.

REMEDY: If the Station Identifier is valid, create an appropriate SCR record in the HTF and warmboot the process.

2008	(2)	UNABLE TO PROCESS PSTM REQUEST. MERCHANT ID FOR FIID AND PSTM RETAILER ID \? NOT FOUND IN \F
-------------	------------	---

where: '\\\ ' specifies the FIID from the PSTM request.

'?' specifies the Retailer Id from the PSTM request.

'F' specifies the HRF file name.

CAUSE: The PSTM request from the Acquirer contains the missing details specified. A record was not found in the HRF.

REMEDY: Create a record in the HRF.

2010	(2)	UNABLE TO DELETE SUSPENDED RECORD
-------------	------------	--

CAUSE: A timeout has occurred whilst awaiting a response back from the Issuer Host and the relevant suspended request record cannot be found.

REMEDY: Contact Operations Personnel.

2012	(2)	NO STATIONS AVAILABLE FOR DPC #
-------------	------------	--

where: '#' specifies the DPC number from the PSTM request.

CAUSE: A request has been formatted to be sent to the Issuer Host, but no stations for the given DPC number are available.

REMEDY: Contact Operations Personnel.

2013	(1)	MESSAGE FAILURE # ON STATION (\S)
-------------	------------	--

where: '#' specifies the message failure number.

'S' specifies the Station symbolic name.

CAUSE: XPNET could not deliver the message to the intended destination.

REMEDY: Contact Operations Personnel.

2014	(1)	STATION (\S) MARKED DOWN - COUNT OF TIMEOUTS EXCEEDS MAXIMUM
-------------	------------	---

where: 'S' specifies the Station symbolic name.

CAUSE: The maximum number of timed out messages has been exceeded.

REMEDY: Contact Operations Personnel.

2020	(2)	RESPONSE FROM (\S) CONTAINS INVALID DATA FOLLOWS:
-------------	------------	--

where: 'S' specifies the Station symbolic name.

CAUSE: A validation on the response code, approval code, Message field or Referral Phone Number field failed on a response message from the Issuer Host. Log message 7100 dumps the invalid message text following this message.

REMEDY: Contact Operations Personnel.

2075 (0) APAC40 MESSAGE RECEIVED FROM (\S)

where: '\S' specifies the Station symbolic name.

CAUSE: Self explanatory.

REMEDY: No action required, information only.

**2087 (0) PSTM RETURNED WITH RTE STAT **

where: '\\ ' specifies the routing status.

CAUSE: A request could not be routed to the host.

REMEDY: No action required, information only.

2090 (1) ERROR WHILE ATTEMPTING TO RETRIEVE ALL TOKEN DATA

CAUSE: Self explanatory.

REMEDY: Contact Operations Personnel if the problem persists.

2090 (0) EXTERNAL MESSAGE RECEIVED FROM (\S) FOLLOWS:

where: '\S' specifies the Station symbolic name.

CAUSE: Self explanatory.

REMEDY: No action required, information only.

2091 (0) CONFIRMATION FAILURE RECEIVED FROM (\S)

where: '\S' specifies the Station symbolic name.

CAUSE: Self explanatory.

REMEDY: No action required, information only.

2092 (0) DISCONNECT FAILURE RECEIVED FROM (\S)

where: '\S' specifies the Station symbolic name.

CAUSE: Self explanatory.

REMEDY: No action required, information only.

2100 (3) \? FILE NAME NOT FOUND IN THE LCONF

where: '\?' specifies the assign name in the LCONF.

CAUSE: A required assign has not been specified in the LCONF. The process will abend.

REMEDY: Add the required assign to the LCONF.

2105 (3) \? PARAM NOT FOUND IN THE LCONF

where: '\?' specifies the parameter in the LCONF.

CAUSE: A required parameter has not been specified in the LCONF. The process will abend.

REMEDY: Add the required parameter to the LCONF.

2107	(3)	INVALID PS-SEG PARAM FOUND IN THE LCONF
-------------	------------	--

CAUSE: Self explanatory.

REMEDY: Correct the parameter and warmboot the process.

2110	(3)	INVALID POS-LN-CUTOVER PARAM FOUND IN THE LCONF
2115	(3)	INVALID POS-LN-CUTOVER PARAM FOUND IN THE LCONF

CAUSE: Self explanatory.

REMEDY: Correct the parameter and warmboot the process.

7010	(1)	WARMBOOT ABORTED – PREVIOUS STATUS RETAINED
-------------	------------	--

CAUSE: A warmboot was not successful. The previous initialisation characteristics will be retained.

REMEDY: There may be an error message prior to this message which caused the warmboot to abort. Check for previous messages. Contact HELP24 if the problem persists.

7011	(0)	WARMBOOT COMPLETE – NEW STATUS IMPLEMENTED
-------------	------------	---

CAUSE: A warmboot has completed successfully.

REMEDY: No action required, information only.

7100	(2)	<DUMP OF EXTERNAL MESSAGE>
-------------	------------	---

CAUSE: This message dumps the external message text received following an invalid message from the Issuer Host.

REMEDY: Contact Operations Personnel.

7101	(2)	NO STATIONS DECLARED IN THE HCF FOR DPC #
-------------	------------	--

where: ‘#’ specifies the DPC number.

CAUSE: No stations are declared for the specified DPC number.

REMEDY: Add stations to the HCF record for the DPC number and warmboot the process.

7102	(1)	HOST INTERFACE NOT REQUIRED
-------------	------------	------------------------------------

CAUSE: There are no DPCs in the HCF for this process.

REMEDY: Add the DPCs to the HCF for this process.

7103	(3)	TOO MANY DPCS DEFINED IN HCF
-------------	------------	-------------------------------------

CAUSE: More than 8 DPCs are defined in the HCF for a given Issuer Host.

REMEDY: Delete the DPCs not required from the HCF.

7106 (3) \? FILE NAME NOT FOUND IN THE LCONF

where: '\?' specifies the assign name in the LCONF.

CAUSE: A required assign has not been specified in the LCONF. The process will abend.

REMEDY: Add the required assign to the LCONF.

7107 (3) \? PARAM NOT FOUND IN THE LCONF

where: '\?' specifies the parameter in the LCONF.

CAUSE: A required parameter has not been specified in the LCONF. The process will abend.

REMEDY: Add the required parameter to the LCONF.

7120 (3) UNABLE TO INIT EXTENDED MEMORY, ERROR #, FOR # REQUESTS

where: '#' specifies the error number.

'#' specifies the number of requests.

CAUSE: The extended memory cannot be initialised for the given number of requests.

REMEDY: Contact HELP24.

7120 (2) FIID VALUE IS MORE THAN 4 CHARACTERS IN PARAM FIIDS-WITH-DPC-HIF-KEY. PARAM = *****

where: '*****' specifies the parameter value.

CAUSE: Self explanatory.

REMEDY: Correct the parameter value in the Logical Network Configuration File (LCONF).

7121 (2) INVALID FIID IN PARAM FIIDS-WITH-DPC-HIF-KEY. PARAM = *****

where: '*****' specifies the parameter value.

CAUSE: Self explanatory.

REMEDY: Correct the parameter value in the Logical Network Configuration File (LCONF).

7122 (3) THE NUMBER OF FIIDS SUPPLIED EXCEEDS THE MAXIMUM ALLOWED. PLEASE CONTACT ACI.

CAUSE: Self explanatory.

REMEDY: Contact HELP24.

7123 (3) THE PARAM DISC^STATUS^TIMER CONTAINS A NON-NUMERIC ITEM

CAUSE: Self explanatory.

REMEDY: Correct the parameter value in the Logical Network Configuration File (LCONF).

7200 (3) NETWORK INITIALISATION FAILED

CAUSE: The process is unable to initialise the network.

REMEDY: Contact HELP24.

7201 (3) LCONF INITIALISATION FAILED - ERROR \E

where: '\E' specifies the error indicating the failure of the LCONF initialisation.

CAUSE: This is either no LCONF or a failure occurred on reading the LCONF.

REMEDY: Check that the given LCONF does exist on the disk. Contact HELP24 if the problem persists.

7202 (3) LOGGER INITIALISATION FAILED

CAUSE: Unable to open the logger process.

REMEDY: Ensure that the template for the logger exists on the correct disk. Contact HELP24 if the problem persists.

7203 (3) HOST INTERFACE INITIALISATION FAILED

CAUSE: The Host Interface may fail to initialise for various reasons.

REMEDY: Check previous log messages for an indication of the reason for failure and, if possible, rectify the problem. Contact HELP24 if the problem persists.

7204 (0) HOST INTERFACE INITIALISATION COMPLETE

CAUSE: The Host Interface process initialisation has completed successfully.

REMEDY: No action required, information only.

7205 (3) PCS INITIALISATION FAILED

CAUSE: Unable to initialise PCS product.

REMEDY: Contact HELP24.

7208 (2) INVALID MESSAGE RECEIVED FOLLOWS:

CAUSE: An invalid message was received from the Issuer Host. Log message 7100 dumps the invalid message text following this message.

REMEDY: Contact Operations Personnel.

7210 (2) LATE RESPONSE OR LATE REQUEST FAILURE FROM (S) FOLLOWS:

where: '\S' specifies the Station symbolic name.

CAUSE: A late response or late request failure has been received.

REMEDY: Contact Operations Personnel if the problem persists.

7211	(2)	INVALID MESSAGE RECEIVED FROM (S) FOLLOWS:
7212	(2)	INVALID MESSAGE RECEIVED FROM (S) FOLLOWS:

where: 'S' specifies the message source symbolic name.

CAUSE: An invalid message type was received. Log message 7100 dumps the invalid message text following this message.

REMEDY: Contact Operations Personnel.

7308	(2)	UNKNOWN MESSAGE FROM REMOTE PROCESS FOLLOWS:
-------------	------------	---

CAUSE: A request message received from a remote process is not a valid message type. Log message 7309 dumps the invalid message text following this message.

REMEDY: Contact Operations Personnel.

7309	(2)	<DUMP OF UNKNOWN MESSAGE FROM REMOTE PROCESS>
-------------	------------	--

CAUSE: This message is generated after log message 7308, giving the dump of the unknown message.

REMEDY: Contact Operations Personnel.

7310	(2)	UNKNOWN DPC NUMBER \\\ FROM REMOTE MESSAGE
-------------	------------	---

where: '\\\' specifies the DPC number in the message.

CAUSE: The DPC number is not specified in the HCF.

REMEDY: Contact Operations Personnel.

7601	(2)	UNABLE TO GET TABLE ENTRY FOR DPC NUMBER #
-------------	------------	---

where: '#' specifies the DPC number in the message.

CAUSE: A record for this DPC number does not exist in the HCF.

REMEDY: Create a record in the HCF and warmboot the process.

7602	(2)	UNABLE TO GET TABLE ENTRY FOR STATION \S
-------------	------------	---

where: 'S' specifies the Station name.

CAUSE: The station is not specified in the HCF record.

REMEDY: Add the station for the given DPC in the HCF record and warmboot the process.

7604	(1)	DPC NUMBER # NOT CONFIGURED - DISPLAYING ALL
-------------	------------	---

where: '#' specifies the DPC number.

CAUSE: The DPC number is not specified in the HCF record.

REMEDY: Add the DPC number to the HCF record and warmboot the process.

7605	(1)	REQUEST DENIED. NO EXTENDED MEMORY AVAILABLE
-------------	------------	---

CAUSE: Unable to process the request because of a lack of extended memory.

REMEDY: Contact Operations Personnel if the problem persists.

7606 (0) HOST PROCESSOR # NOW MARKED UP

where:	'#' specifies the DPC number.
CAUSE:	A LOGON command has been issued.
REMEDY:	No action required, information only.

7607 (0) HOST PROCESSOR # NOW MARKED DOWN

where:	'#' specifies the DPC number.
CAUSE:	A LOGOFF command has been issued.
REMEDY:	No action required, information only.

7608 (0) STATION \S STARTED

where:	‘S’ specifies the Station number.
CAUSE:	A station has been started.
REMEDY:	No action required, information only.

7609 (0) STATION \S STOPPED

where: '\S' specifies the Station number.

CAUSE: A station has been stopped.

REMEDY: No action required, information only.

7610 (2) UNABLE TO WRITE TO HTF, KEY //////////////////////////////////////, ERROR #

where:	'\'' specifies the HTF record key.
	'#' specifies the error number.
CAUSE:	An error occurred on updating the HTF.
REMEDY:	Contact Operations Personnel.

B. MESSAGE FORMATS

The abbreviations used in this appendix are:

AN	Alphanumeric value
F	Fixed value
FS	Field Separator
M	Mandatory field
N	Numeric value
O	Optional field
US	Unit Separator
V	Variable value

B.1 AUTHORISATION REQUEST

FIELD					FIELD NAME
No.	Size	F or V	M or O	Range	
0	1	F	M	N	Dial Indicator
1	8	F	M	AN	Terminal Identification
2	4	F	M	N	Message Number
3	4	F	M	N	Terminal Type
4	2	F	M	N	Message Type Identification
5	15	V	M	AN	Merchant Number
6	1	F	M	FS	FS
7	40	V	M	AN	Card Details
8	1	F	M	FS	FS
9	11	V	M	N	Amount
10	1	F	M	FS	FS

B.2 AUTHORISATION RESPONSE

FIELD					FIELD NAME
No.	Size	F or V	M or O	Range	
0	1	F	M	N	Dial Indicator
1	8	F	M	AN	Terminal Identification
2	4	F	M	N	Message Number
3	2	F	M	N	Message Type Identification
4	2	F	M	N	Response Code
5	1	F	M	N	Confirmation Request
6*	9	V	O	N	Authorisation Code
7	1	F	M	FS	FS
8*	11	V	O	N	Amount
9	1	F	M	FS	FS
10	80	V	M	AN	Message
11Δ	1	F	M	FS	FS
12*	16	V	O	N, US	Referral Telephone Number
13	1	F	O	FS	FS
14*	3	V	O	N	Floor Limit (Pre-communications)
15	1	F	O	FS	FS
16*~	4	F	O	N	Date
17	1	F	O	FS	FS

* These fields are optional and may be omitted. If any subsequent fields are used, the field delimiter (FS) for an omitted field must be included.

Δ The response message may be terminated after field 11.

~ The terminal must not mis-operate if up to 12 digits are sent in Field 16, as per EFT response field.

B.3 AUTHORISATION CONFIRMATION

FIELD					FIELD NAME
No.	Size	F or V	M or O	Range	
0	1	F	M	N	Dial Indicator
1	8	F	M	AN	Terminal Identification
2	4	F	M	N	Message Number
3	4	F	M	N	Terminal Type
4	2	F	M	N	Message Type Identification
5	1	F	M	N	Confirmation Code

B.4 HOLD MESSAGE

FIELD					FIELD NAME
No.	Size	F or V	M or O	Range	
0	1	F	M	N	Dial Indicator
1	8	F	M	AN	Terminal Identification
2	4	F	M	N	Message Number
3*	2	F	M	N	Message Type Identification
4	2	F	M	N	Response Code
5	1	F	M	N	Confirmation Request
6	9	V	O	N	Authorisation Code
7	1	F	M	FS	FS
8	11	V	O	N	Amount
9	1	F	M	FS	FS
10	80	V	M	AN	Message
11	1	F	M	FS	FS

- * These hold message may be terminated with the ETX and LRC after field 3, if no message is to be sent.

There is no limit to the number of hold messages with a zero response code that can be sent during a transaction.

C. SAMPLE COMMUNICATIONS ENVIRONMENT & CONFIGURATION

C.1 X25AM BIT SYNC LINE (PSS)

12protocol, circuits, DTE, PSS, RSIZE, PACKETSIZE, 12window set by BT.

C.1.1 PSS^MACRO

<i>X25AM</i>	
<i>TYPE 61 ,</i>	<i>SUBTYPE 00,</i>
<i>INTERRUPT</i>	<i>AXCOM6203INTERRUPT ,</i>
<i>DRIVER</i>	<i>AXCOM6203FDX ,</i>
<i>L2PROTOCOL</i>	<i>X25^LAPB^PROTOCOL ,</i>
<i>L4PROTOCOL1</i>	<i>X25^PTP^PROTOCOL ,</i>
<i>L4PROTOCOL0</i>	<i>X25^ITI^PROTOCOL ,</i>
<i>CIRCUITS</i>	<i>16 ,</i>
<i>ASCII ,</i>	<i>DTE ,</i>
<i>PSS ,</i>	<i>STARTDOWN ,</i>
<i>RSIZE</i>	<i>128 ,</i>
<i>PACKETSIZE</i>	<i>255 ,</i>
<i>L2WINDOW</i>	<i>2 ,</i>
<i>L3WINDOW</i>	<i>3 #;</i>

C.1.2 Line

<i>OBJECT</i>	<i>LINE \ACIUK.\$PSS</i>
<i>CALLINGADDR</i>	<i>NETADDR</i>
<i>CALLS</i>	<i>TWOWAY</i>
<i>CHARACTERSET</i>	<i>ASCII CUG</i>
<i>DSRTIMEOUT</i>	<i>20.00</i>
<i>EXTENDED</i>	<i>OFF</i>
<i>FRAMEMODE</i>	<i>DTE</i>
<i>IDLETIMEOUT</i>	<i>15.00</i>
<i>INTERFACE</i>	<i>RS232</i>
<i>NETID</i>	<i>PSS</i>
<i>NETADDR</i>	<i>234272700102</i>
<i>BPADPARMS</i>	
<i>CPADPARMS</i>	
<i>NOBUFFRETRY</i>	<i>152</i>
<i>NOBUFFTIMER</i>	<i>0.00</i>
<i>RNR</i>	<i>OFF</i>
<i>PROGRAM</i>	<i>C9311P00</i>
<i>PVC</i>	
<i>RETRIES</i>	<i>10</i>
<i>SVC</i>	<i>(1024, 1039)</i>
<i>SYNCS</i>	<i>3</i>
<i>THRESHOLD</i>	<i>500</i>
<i>TIMEOUT</i>	<i>3.00</i>
<i>L3MODULUS</i>	<i>8</i>
<i>L3WINDOW</i>	<i>3</i>
<i>DEBUG</i>	<i>OFF</i>
<i>NAMES</i>	<i>#HERTZ</i>
<i>TYPE</i>	<i>(61, 0)</i>

Note that '#Hertz' is the subdevice used by the Host Interface.

C.1.2.1 Subdevice used by Host Interface

<i>OBJECT</i>	<i>LU \ACIUK.\$PSS.#HERTZ</i>
<i>ACCEPTCHG</i>	<i>OFF</i>
<i>ASSOCIATE</i>	
<i>CUG</i>	
<i>NEGOTIATE</i>	
<i>L4TIMEOUT</i>	<i>20.00</i>
<i>NETADDR</i>	<i>234262100157</i>
<i>NULLFILL</i>	<i>OFF</i>
<i>PARITY</i>	<i>NONE</i>
<i>PARITYCHK</i>	<i>OFF</i>
<i>PRICALL</i>	<i>OFF</i>
<i>REMOTENET</i>	<i>PSS</i>
<i>PORT</i>	<i>5</i>
<i>PROTOCOL</i>	<i>PTP</i>
<i>PVC</i>	
<i>RECSIZE</i>	<i>128</i>
<i>REVERSECHG</i>	<i>OFF</i>
<i>THRUPUT</i>	<i>(0, 0)</i>
<i>TYPE</i>	<i>(9, 0)</i>

Note that a subdevice is required for each station declared in the NEFS. The *NETADDR* values to be configured for the subdevice will be as specified by the issuer system.

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D. STANDARD TRANSLATION TABLE

EXTERNAL RESPONSE CODE		INTERNAL RESPONSE CODE	
Code	Description	Code	Description
00	Approval	000	Approved with balances
01	Refer to card issuer	101	Issue a call
02	Refer to card issuer	101	Issue a call
05	Do not honour	050	Declined
08	Honour with Id	004	Approved with pending identification (signed paper draft required)
11	Approve VIP	006	Approved with VIP status
21	No action taken	101	Issue a call
30	Format error	069	Bad message edit
33	Expired card	051	Expired card
34	Suspected fraud	057	Lost or stolen card status
36	Restricted card	059	Restricted status
38	PIN tries exceeded	052	PIN tries exceeded
41	Lost card	057	Lost or stolen card status
43	Stolen card	057	Lost or stolen card status
51	Insufficient funds	076	Insufficient funds
54	Expired card	051	Expired card
55	Incorrect PIN	201	Incorrect PIN
56	No card account	060	Account not on the CAF
59	Suspected fraud	057	Lost or stolen card status
62	Restricted card	069	Bad message edit
63	Security violation	101	Issue a call
94	Duplicate transmission	078	Duplicate transaction

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E. GLOSSARY

Assign	The designating of a resource, for example, a file that will be used by a program during execution.
Data Processing Centre (DPC)	A logical channel between the front-end issuer/acquirer system and a back-end acquirer/issuer host.
Issuer Host	An external host which performs authorisation for transactions acquired by a front-end system.
Issuer Host Interface	An interface which connects a front-end acquirer system to a back-end issuer host.
Logical Network	A network of systems that share the same application database.
Network Management Messages	Messages used to control the communication channels between the front-end and back-end systems.
Parameter	A value used in processing that controls how processing is performed or the results produced and which can be changed to affect processing or results.
Performance	The capability of devices and systems in terms of speed and reliability.
Primary Key	The main or only value by which a record is identified for access purposes.
Protocol	A set of rules for performing data exchanges between independent devices. Protocols may exist at many levels in a network such as link-by-link and end-to-end.
Response Time	The period between the completion of message input at the source and the start of message output at the destination.
Station	A logical subchannel associated with a DPC. A station can be configured to receive transactions only or to send and receive transactions.
Status	The condition a device or operation is in at a particular time.
Store and Forward	A file containing completions or advices to be sent to a back-end issuer host.

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Release 5.1

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☐☐☐

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